

Isobutane-n-Butane Distillation Column Design

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Parameter	Value
System	Isobutane / n-Butane
F	100.00 kmol/h
zF	0.50 mol/mol
xD	0.85 mol/mol
xB	0.45 mol/mol
P	1.00 atm
q	1.00
T	-9.82 °C
α	1.8920
Eo	0.50
Tray spacing	0.50 m
Rmin	1.2694
R (optimum)	1.5381

Operating Lines

Rectifying line: $y = 0.6060x + 0.3349$

q-line: $x = 0.5$

Stripping line: $y = 3.7579x + -1.2411$

N_theoretical: 7 stages

N_actual: 15 trays

Stage-by-Stage Compositions

Stage	x [mol/mol]	y [mol/mol]
0	0.850000	0.850000 (starting point)
1	0.749688	0.789210
2	0.664300	0.737464
3	0.597528	0.696999
4	0.548694	0.667406
5	0.514701	0.646805
6	0.491844	0.607247
7	0.449696	0.448859

Column Sizing

Parameter	Value
Column height H_{vessel}	9.50 m
Vapor flow V	31.73 kmol/h
Column diameter D_c	0.800 m
Column cross-sectional area A_{col}	0.502 m ²

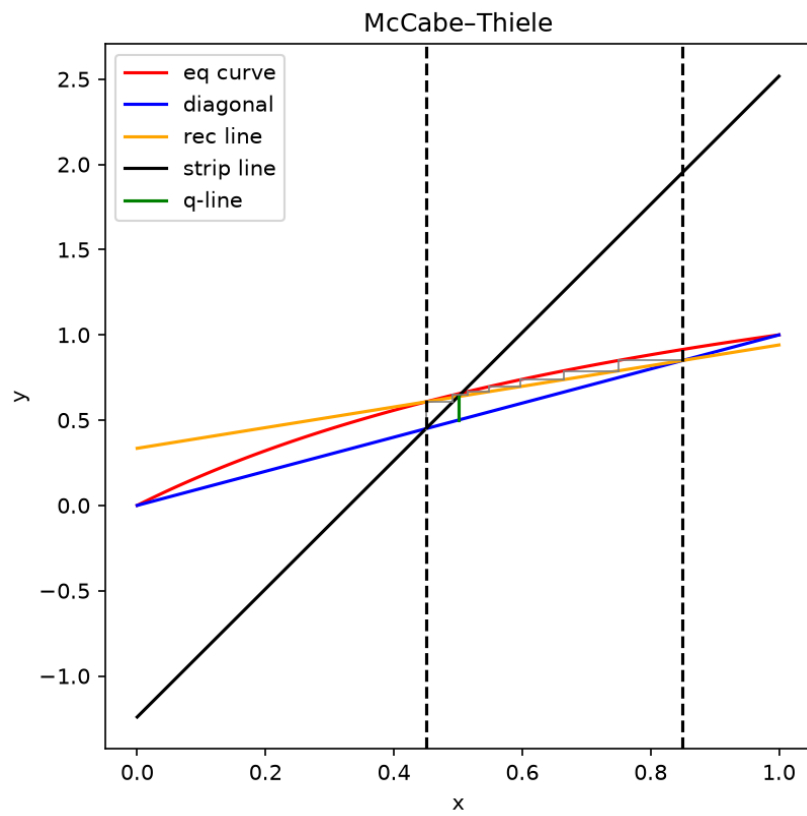
Energy Duties

Parameter	Value
Condenser duty Q_c	1.110e+06 kJ/h
Reboiler duty Q_r	1.110e+06 kJ/h

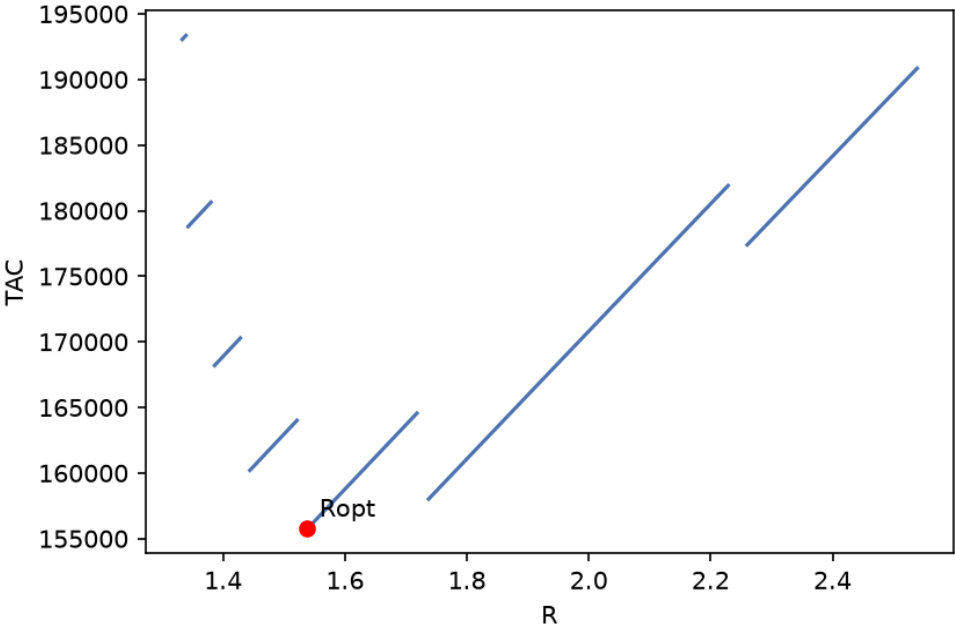
Costs

Parameter	Value
Steam cost	\$106,602 / year
Cooling cost	\$3,109 / year
Shell cost	\$37,882
Tray cost	\$137,094
Condenser cost	\$49,792
Reboiler cost	\$58,090
Total purchase cost	\$282,858
Capital recovery factor (CRF)	0.16275 1/year
Annualized capital cost	\$46,034 / year
Total annual cost (TAC)	\$155,745 / year

McCabe-Thiele Diagram



TAC vs Reflux Ratio



Economic Breakdown

